

Salauddin Tapu

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Github: github.com/salauddintapu

Google Scholar Profile

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EDUCATION

- Gachon University** Seongnam, South Korea
• *M.S. in Chemical and Biological Engineering* Mar 2026 – Present
 - Research Interests: Soft Robotics, UAV, Artificial Intelligence
 - Thesis: **To be determined**
- Khulna University** Khulna, Bangladesh
• *B.Sc. in Electronics and Communication Engineering CGPA: 3.35/4.00* Jan 2018 – Mar 2023
 - Research Interests: Machine Learning
 - Thesis: **Prediction of Arrhythmia after Acute Myocardial Infarction Using Machine Learning and Statistical Techniques**

PUBLICATIONS

- M. S. Chowdhury, **S. Tapu**, N. K. Sarkar, F. B. Ali, L. Sabrin, *et al.*, “Med-2D SegNet: A Lightweight Deep Neural Network for Medical 2D Image Segmentation,” *arXiv preprint*, 2025.
- U. U. Gronthy, U. Biswas, **S. Tapu**, M. A. Samad, and A.-A. Nahid, “A bibliometric analysis on arrhythmia detection and classification from 2005 to 2022,” *Diagnostics*, vol. 13, no. 10, p. 1732, 2023.

ACADEMIC RESEARCH EXPERIENCE AND PUBLICATIONS

- Graduate Research Assitant:** Prediction of Arrhythmia after Acute Myocardial Infarction Using Machine Learning and Statistical Techniques, Dataset: AMI Dataset

SKILLS SUMMARY

- Languages:** C, Python
- Tools:** Git, Scikit-learn, Numpy, Pandas, OpenCV, Flask, TensorFlow, Keras, PyTorch, Colab, Kaggle

WORK EXPERIENCE

- Anwar Technologies** Dhaka, Bangladesh
• *Junior Machine Learning Engineer* Nov 2023 - Nov 2025
 - Bangla Handwritten OCR System (Research Production):**
 - The goal of the research project was to create an advanced Optical Character Recognition (OCR) system capable of precisely identifying handwritten Bangla text at both the word and sentence levels.
 - We incorporate semi-supervised learning with deep neural networks to optimize data creation and annotation. Our OCR model, built on a CRNN architecture with a ResNet34 backbone, is trained on both labeled and unlabeled data to improve accuracy and adaptability. To further enhance performance, we generated synthetic text images with diverse backgrounds and applied various augmentation techniques, enabling the model to generalize more effectively across different text and image variations.
 - Our research advances OCR technology by tackling the challenges specific to recognizing handwritten Bangla text. Through the use of semi-supervised learning, we address issues of data scarcity and limited annotations, creating scalable and cost-effective OCR solutions designed for under-resourced languages and scripts.
 - This research project led to the creation of an innovative OCR system capable of accurately detecting and transcribing handwritten Bangla text. By utilizing semi-supervised learning techniques, we significantly enhanced recognition accuracy and data efficiency, showcasing the potential for scalable and adaptable OCR solutions in multilingual environments.
 - OCR Project:**
 - Document Classifier:** Developed a document classification system to automatically categorize different document types before performing OCR which enhanced processing efficiency by reducing OCR errors and optimizing document-specific text recognition.
 - Document Parsing:** Integrated LLM-based parsing to extract structured information from various government documents, ensuring accurate data retrieval. Addressed challenges like inconsistent document formats and text distortions through preprocessing techniques.
 - LLM Chatbot:** Designed and implemented a Retrieval-Augmented Generation (RAG) chatbot leveraging LLaMA 3.1 for enhanced contextual understanding and accurate responses. Developed an efficient retrieval pipeline to fetch relevant documents and fine-tuned the model for domain-specific queries. Optimized prompt engineering and response generation to improve accuracy and coherence. Integrated the chatbot into a user-friendly interface, ensuring seamless interaction and real-time information retrieval.
 - Face Recognition System:** Developed a system using Raspberry Pi with a camera to capture video feed and stream all image frames to a server for processing. Implemented advanced preprocessing techniques, including noise removal, resizing, and brightness/contrast adjustment, to enhance image quality. Designed and trained high-precision face detection and recognition models, achieving accurate real-time recognition performance.

- Anwar Technologies** Dhaka, Bangladesh
Machine Learning Engineer (Intern) Aug 2023 - Nov 2023
 - OCR System:** Developed an OCR system to automate document digitization for vendor management and resume sorting, involving data collection from diverse documents (TIN, NID, CVs, BIN, etc.), preprocessing with noise removal and labeling, and designing data pipelines to process user inputs. The system incorporated document analysis to identify patterns for improved accuracy and included a model training pipeline for detection and information extraction.
- VISIE Limited** Dhaka, Bangladesh
Junior Machine Learning Engineer (Project Based) Jan 2023 - Dec 2024
 - eKYC Systems:** Built end-to-end eKYC systems, including Passport KYC for secure passport validation with fraud prevention and NID eKYC with advanced face matching for fast, reliable identity verification.
 - Face Recognition System:** Developed a high-speed, real-time attendance system using live CCTV footage, powered by advanced face recognition technology that's fast, reliable, and built to perform seamlessly in any environment.
 - FinTech Project:** We developed an OCR system incorporating YOLOv5 to process and analyze Credit Information Bureau (CIB) reports and bank statement documents efficiently.
 - visie.tech

LANGUAGE PROFICIENCY

- IELTS:** Overall Band Score - 7.5 (*May 2025*)
 - Listening: 7.0 — Reading: 8.0 — Writing: 7.5 — Speaking: 6.5

VOLUNTEER WORKS

- President — Khulna University Career Club** Khulna, Bangladesh
Led 15+ events, including 1 national and 1 regional, with a team of 50+ members Apr 2022 - Feb 2023
- International Service Director — Rotaract Club of Khulna University** Khulna, Bangladesh
Worked and organized 6+ events, worked in a team of 30+ members Jul 2020 - Apr 2021

REFERENCE

- Prof. Jongkuk Ko**
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- Prof. Dr. Uzzal Biswas**
 Electronics and Communication Engineering Discipline, Khulna University
 Email: uzzal.biswas@ece.ku.ac.bd